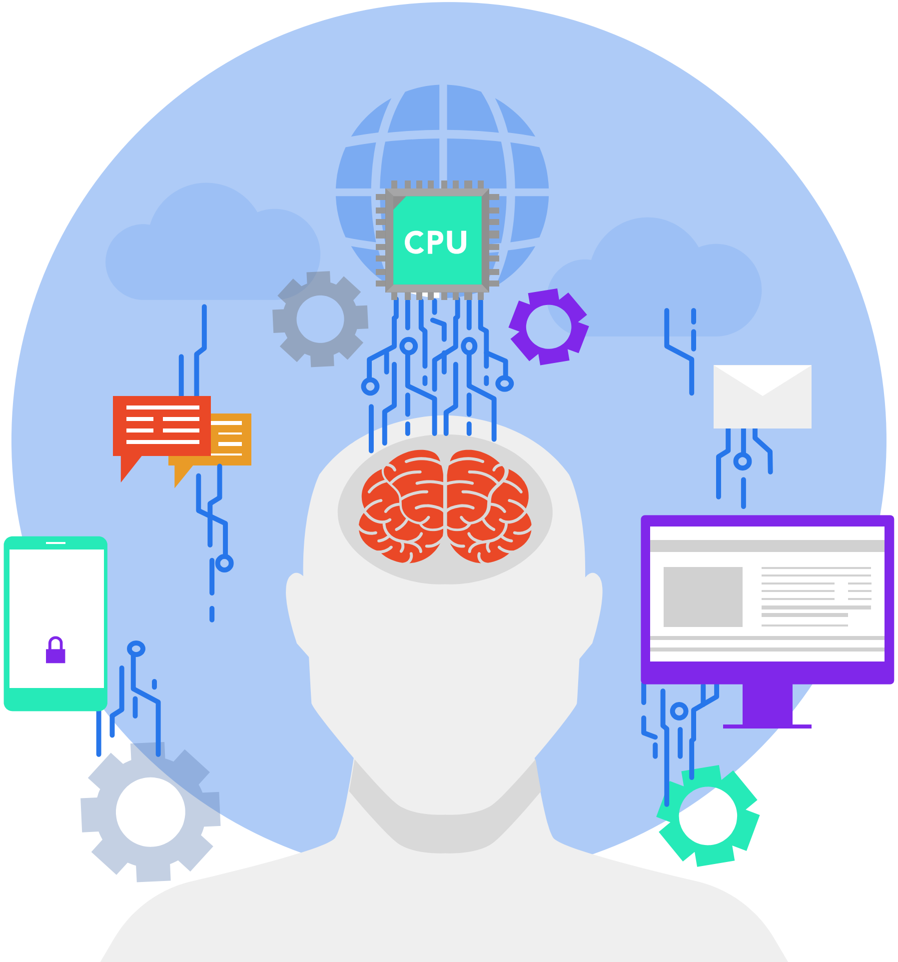


Support Vector Machine

Diyah Utami Kusumaning Putri, S.Kom., M.Sc., M.Cs.
Afiahayati, S.Kom., M.Cs., Ph.D.
Department of Computer Science and Electronics
Universitas Gadjah Mada
diyah.utami.k@ugm.ac.id
afia@ugm.ac.id

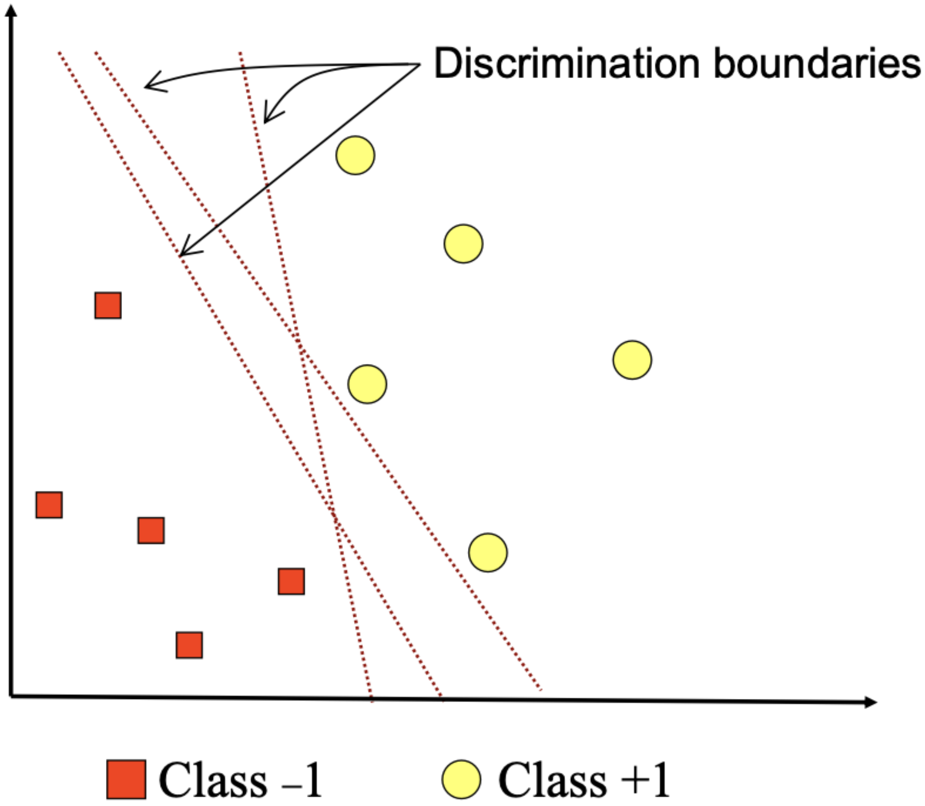
Support Vector Machine

- The Support Vector Machine (SVM) was developed by Boser, Guyon, Vapnik, and was first presented in 1992 at the Annual Workshop on Computational Learning Theory.
- The basic concept of SVM is actually a harmonious combination of computational theories that have existed decades before, such as the hyperplane margin (Duda & Hart in 1973, Cover 1965, Vapnik 1964, etc.), the kernel introduced by Aronszajn in 1950, and other supporting concepts.
- As a pattern recognition method, SVM is still relatively young. However, evaluation of its capabilities in its various applications places it as a state of the art in pattern recognition and is currently one of the fastest-growing themes.

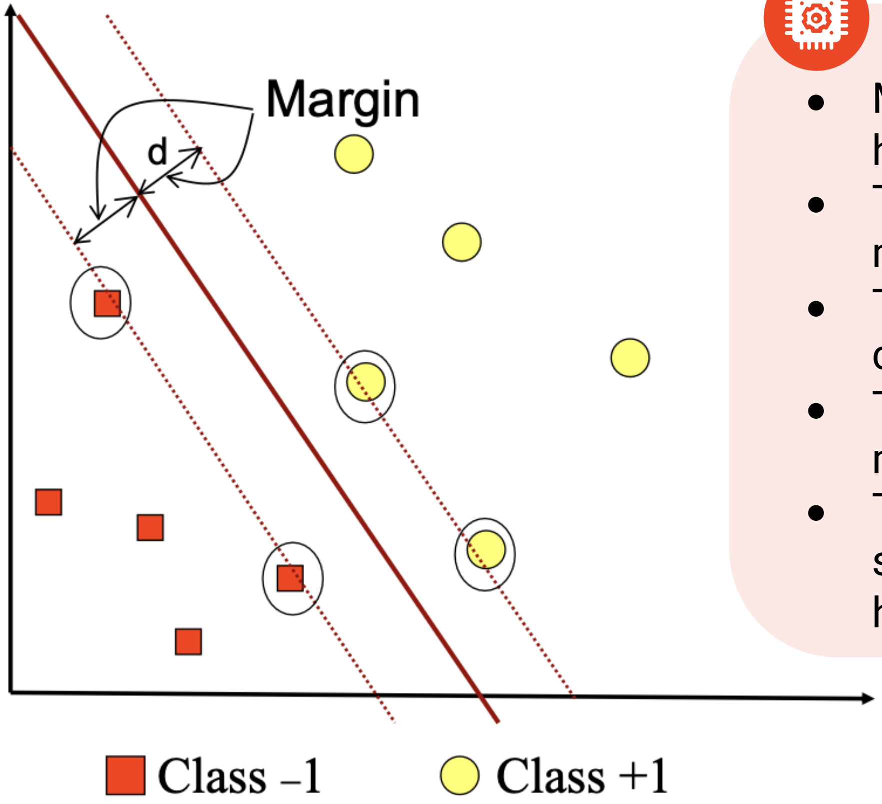
Support Vector Machine

- In contrast to the neural network strategy which seeks to find a separating hyperplane between classes, SVM is a machine learning method that works on the principle of Structural Risk Minimization (SRM) with the aim of finding the best hyperplane that separates two classes in the input space.
- The basic principle of SVM is a linear classifier and further developed to work on non-linear problems, by incorporating the concept of a kernel trick in a high-dimensional workspace.
- This development stimulates interest in research in the field of pattern recognition to investigate the potential of SVM theoretically and in terms of application. Nowadays, SVM has been successfully applied to real-world problems, and in general, it provides better solutions than conventional methods such as ANN.

Binary Classification

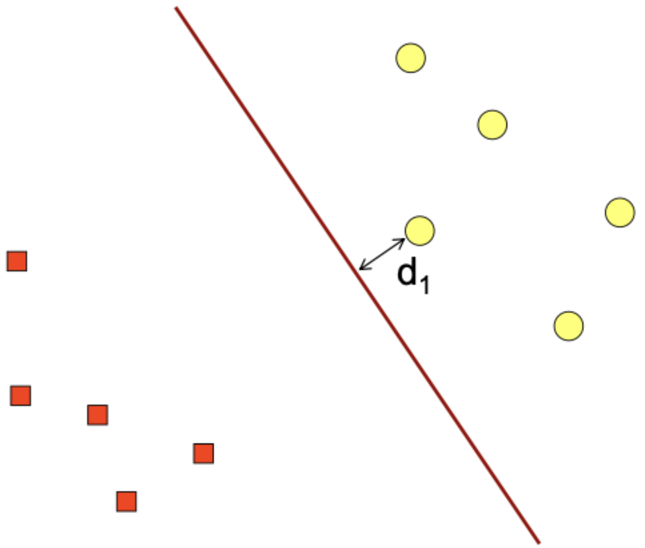


Optimal Hyperplane by SVM

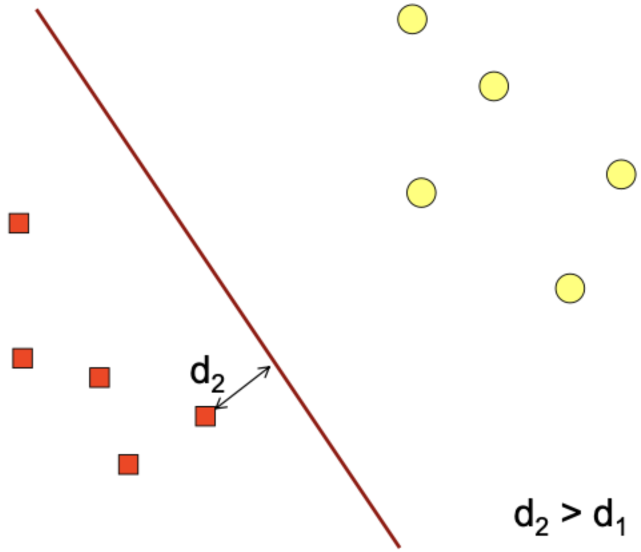


- Margin (d) = minimum distance between hyperplane and training samples.
- The best hyperplane is obtained by maximizing the value of the margin
- The sample closest to the hyperplane is called the support vector.
- The best hyperplane is located right in the middle of the two classes.
- The learning process in SVM: looking for support vectors to get the optimal hyperplane.

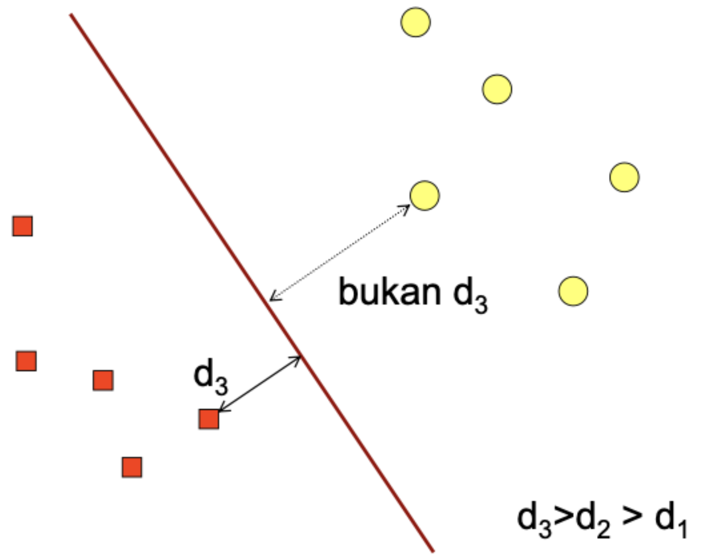
Optimal Hyperplane by SVM



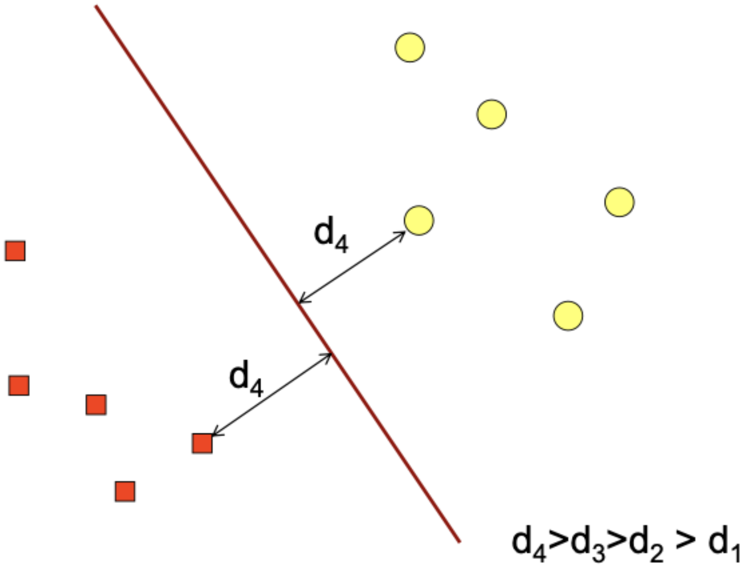
Optimal Hyperplane by SVM



Optimal Hyperplane by SVM



Optimal Hyperplane by SVM

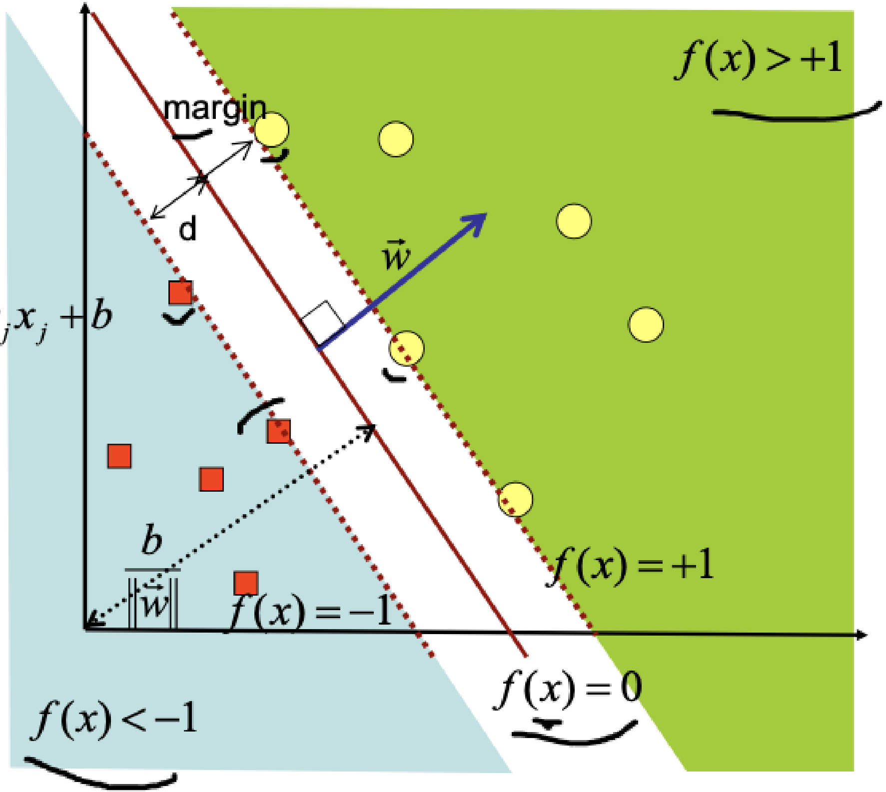


Separating Hyperplane for 2D

$$\vec{x} = (x_1, x_2, \cdots, x_d)^T$$

$$f(x) = < \vec{w} \cdot \vec{x} > + b = \sum_{j=1}^{\dim} w_j x_j + b \quad (1)$$

bias



Optimal Hyperplane by SVM

- Margin (d) = minimum distance between hyperplane and training samples.
- The best hyperplane is obtained by maximizing the value of the margin
- How to maximize d?

Training set: $(\vec{x}_1, y_1), (\vec{x}_2, y_2), \cdots, (\vec{x}_l, y_l)$

pattern ($\vec{x}_1$)   class-label (+1 atau -1) (y_1)

- Distance between hyperplane and x pattern on the training set.

$$\frac{\left| < \vec{w} \cdot \vec{x} > + b \right|}{\left\| \vec{w} \right\|} \quad (2)$$

- The minimum distance between the hyperplane and training set.

$$\min_{i=1,\ldots,l} \frac{\left| < \vec{w} \cdot \vec{x}_i > + b \right|}{\left\| \vec{w} \right\|} \quad (3)$$

Optimal Hyperplane by SVM

Constraint : $$\min_{i=1,\ldots,l} \left| < \vec{w} \cdot \vec{x}_i > + b \right| = 1 \qquad (4)$$

Substitution (4) to (3) is obtained, then the distance between hyperplane and training set becomes

must be maximized $$\frac{1}{\|\vec{w}\|} \qquad (5)$$

PRIMAL FORM

Minimize $$\|\vec{w}\|^2 \qquad (6)$$

Subject to $$y_i(< \vec{w} \cdot \vec{x}_i > + b) \geq 1 \quad (i = 1,2,\ldots,l) \qquad (7)$$

(7): data is assumed to be 100% correctly classified

Optimal Hyperplane by SVM

The Langrange Multiplier is used to simplify (6) and (7) becomes

$$L(\vec{w}, b, \alpha) = \frac{1}{2}\|\vec{w}\|^2 - \sum_{i=1}^{l} \alpha_i (y_i (< \vec{x}_i \cdot \vec{w} > +b) - 1) \qquad (8)$$

where $\alpha_i \geq 0$ (9)

The solution can be obtained by minimizing L to w, b (primal variables) and maximizing L to α_i (dual variables).
When the solution is obtained (optimal point), the gradient $L = 0$

Therefore

$$\frac{\partial L}{\partial b} = 0, \frac{\partial L}{\partial \vec{w}} = 0 \qquad (10)$$

So that it is obtained

$$\sum_{i=1}^{l} \alpha_i y_i = 0 \qquad (11)$$

$$\vec{w} = \sum_{i=1}^{l} \alpha_i y_i \vec{x}_i \qquad (12)$$

Optimal Hyperplane by SVM

(11) and (12) are substituted into (8) so that we get (13) and (14)

$$\sum_{i=1}^{l} \alpha_i y_i = 0 \quad (11)$$

$$\vec{w} = \sum_{i=1}^{l} \alpha_i y_i \vec{x}_i \quad (12)$$

$$L(\vec{w}, b, \alpha) = \frac{1}{2}\|\vec{w}\|^2 - \sum_{i=1}^{l} \alpha_i (y_i (< \vec{x}_i \cdot \vec{w} > + b) - 1) \quad (8)$$

The obtained function maximizes only one α variable

Maximize α

$$\sum_{i=1}^{l} \alpha_i - \frac{1}{2} \sum_{i,j=1}^{l} \alpha_i \alpha_j y_i y_j < \vec{x}_i \cdot \vec{x}_j > \quad (13)$$

Subject to

$$\alpha_i \geq 0 \quad (i = 1, 2, \ldots, l) \qquad \sum_{i=1}^{l} \alpha_i y_i = 0 \quad (14)$$

Optimal Hyperplane by SVM

$$L(\vec{w},b,\alpha) = \frac{1}{2}\|\vec{w}\|^2 - \sum_{i=1}^{l}\alpha_i(y_i(<\vec{x}_i\cdot\vec{w}>+b)-1)$$

$$\|\vec{w}\|^2 = <\vec{w}\cdot\vec{w}>$$

$$= \left\langle \sum_{i=1}^{l}\alpha_i y_i \vec{x}_i \cdot \sum_{j=1}^{l}\alpha_j y_j \vec{x}_j \right\rangle$$

$$= \boxed{\sum_{i,j=1}^{l}\alpha_i\alpha_j y_i y_j <\vec{x}_i\cdot\vec{x}_j>} \quad \mathbf{A}$$

$$\sum_{i=1}^{l}\alpha_i(y_i(<\vec{x}_i\cdot\vec{w}>+b)-1) = \sum_{i=1}^{l}\alpha_i y_i <\vec{x}_i\cdot\vec{w}> + \sum_{i=1}^{l}\alpha_i y_i b - \sum_{i=1}^{l}\alpha_i$$

$$= \sum_{i=1}^{l}\alpha_i y_i \left\langle \vec{x}_i \cdot \sum_{j=1}^{l}\alpha_j y_j \vec{x}_j \right\rangle + 0 - \sum_{i=1}^{l}\alpha_i$$

$$= \underbrace{\sum_{i=1}^{l}\alpha_i\alpha_j y_i y_j \langle\vec{x}_i\cdot\vec{x}_j\rangle}_{\mathbf{A}} + 0 - \boxed{\sum_{i=1}^{l}\alpha_i} \quad \mathbf{B}$$

Optimal Hyperplane by SVM

½ A – (A-B) = B- ½ A

$$L(\vec{w},b,\alpha) = \frac{1}{2}\|\vec{w}\|^2 - \sum_{i=1}^{l}\alpha_i(y_i(<\vec{x}_i\cdot\vec{w}>+b)-1)$$

$$\begin{aligned}
\|\vec{w}\|^2 &= <\vec{w}\cdot\vec{w}> \\
&= \left\langle \sum_{i=1}^{l}\alpha_i y_i \vec{x}_i \cdot \sum_{j=1}^{l}\alpha_j y_j \vec{x}_j \right\rangle \\
&= \boxed{\sum_{i,j=1}^{l}\alpha_i\alpha_j y_i y_j <\vec{x}_i\cdot\vec{x}_j>}
\end{aligned}$$

A

$$\begin{aligned}
\sum_{i=1}^{l}\alpha_i(y_i(<\vec{x}_i\cdot\vec{w}>+b)-1) &= \sum_{i=1}^{l}\alpha_i y_i <\vec{x}_i\cdot\vec{w}> + \sum_{i=1}^{l}\alpha_i y_i b - \sum_{i=1}^{l}\alpha_i \\
&= \sum_{i=1}^{l}\alpha_i y_i\left\langle \vec{x}_i\cdot\sum_{j=1}^{l}\alpha_j y_j\vec{x}_j\right\rangle + 0 - \sum_{i=1}^{l}\alpha_i \\
&= \boxed{\sum_{i=1}^{l}\alpha_i\alpha_j y_i y_j\langle\vec{x}_i\cdot\vec{x}_j\rangle} + 0 - \boxed{\sum_{i=1}^{l}\alpha_i}
\end{aligned}$$

A

B

Optimal Hyperplane by SVM

$$L(\vec{w}, b, \alpha) \quad = \quad \frac{1}{2}\left\|\vec{w}\right\|^2 - \sum_{i=1}^{l} \alpha_i (y_i(< \vec{x}_i \cdot \vec{w} > +b) - 1)$$

$$L(\vec{w}, b, \alpha) \quad = \quad \sum_{i=1}^{l} \alpha_i - \sum_{i=1}^{l} \alpha_i \alpha_j y_i y_j \left\langle \vec{x}_i \cdot \vec{x}_j \right\rangle$$

Optimal Hyperplane by SVM

DUAL FORM

The obtained function maximizes only one α variable

Maximize α

$$\sum_{i=1}^{l}\alpha_i - \frac{1}{2}\sum_{i,j=1}^{l}\alpha_i\alpha_j y_i y_j < \vec{x}_i \cdot \vec{x}_j > \quad (13)$$

Subject to

$$\alpha_i \geq 0 \quad (i = 1,2,\ldots,l) \qquad \sum_{i=1}^{l}\alpha_i y_i = 0 \quad (14)$$

The above formula is a Quadratic Programming problem, whose solution α_i is mostly 0 .
The data x_i from the training set where α_i is not 0 (positive value) is called the support vector (the most informative part of the training set).

The training process in SVM is aimed at finding the α_i value

Optimal Hyperplane by SVM

If α has been obtained, then $\vec{w}$ and b can be obtained as follows:

$$\vec{w} = \sum_{i=1}^{l} \alpha_i y_i \vec{x}_i \qquad (12)$$

$$b = -\frac{1}{2}\left(< \vec{w} \cdot \vec{x}_{-1} > + < \vec{w} \cdot \vec{x}_{+1} >\right) \qquad (15)$$

The classification of $\vec{t}$ pattern is calculated as follows:

$$f(\vec{t}) = \operatorname{sgn}\left(\sum_{i=1, x_i \in SV}^{n} \alpha_i y_i < \vec{t} \cdot \vec{x}_i > + b \right) \qquad (16)$$

Note

There are two important things to remember:

- Equation (13) only has a single global maximum that can be calculated efficiently
- The data is not viewed individually but in the form of a dot product of two pieces of data

Hard vs Soft Margin

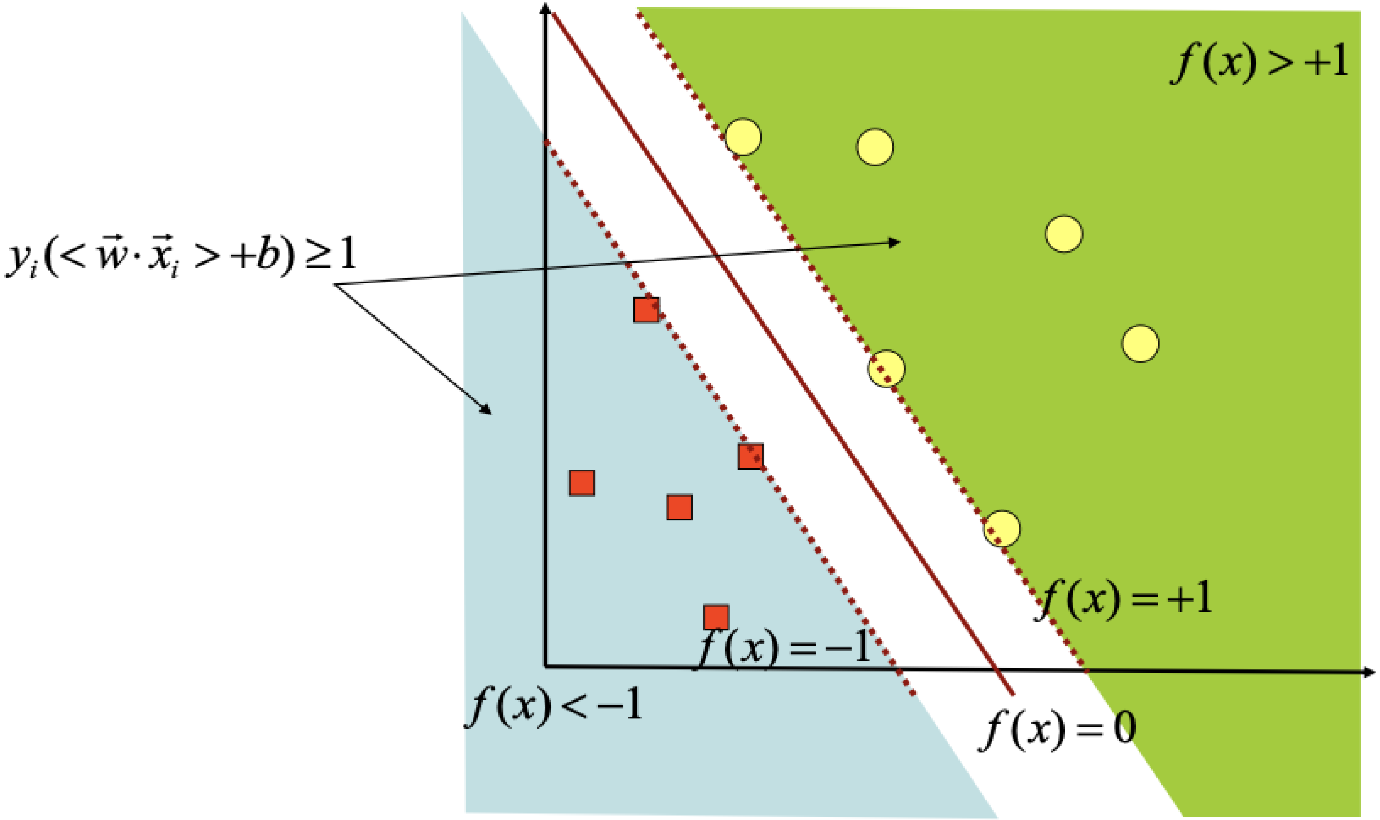
Minimize	$\lVert \vec{w} \rVert^2$	(6)
Subject to	$y_i(<\vec{w}\cdot\vec{x}_i> +b) \geq 1 \quad (i = 1,2,\ldots,l)$	(7)

In the previous calculation, according to equation (7), the data is assumed to be 100% classified correctly (hard margin). In fact, that is not true.

Generally, the data cannot be classified as 100% correct, so the above assumptions cannot apply and a solution cannot be found.

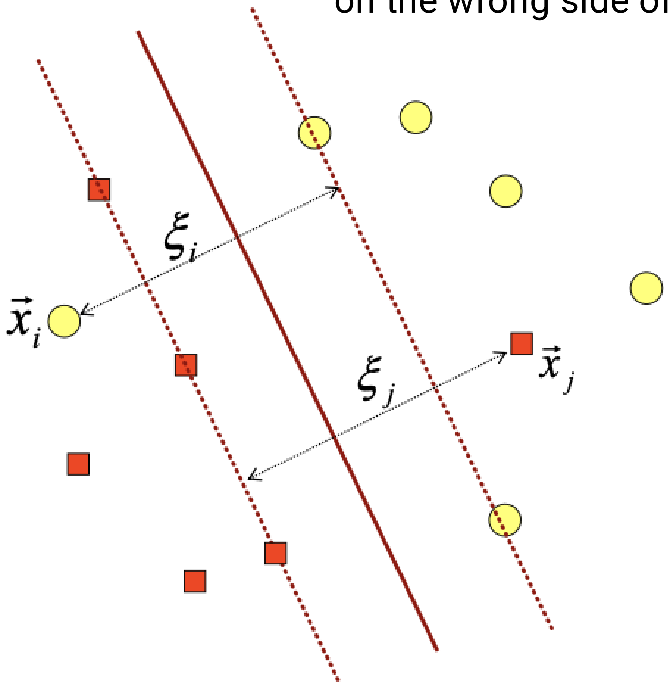
To solve this problem, SVM was reformulated by introducing the soft margin technique.
Soft margin: softens the constraint by providing a perfect tolerance for unclassified data.

Separating Hyperplane for 2D



Separating Hyperplane for 2D

$\xi > 1$ indicates that the training example is on the wrong side of the hyperplane



Soft Margin

Soft margin is obtained by entering the slack variable $\xi_i (\xi_i \geq 0)$ into equation (7), so that

$$y_i(<\vec{w}\cdot\vec{x}_i> + b) \geq 1 - \xi_i \quad (i = 1,2,\ldots,l) \tag{17}$$

While the optimized objective function (6) becomes

$$\text{minimize} \quad \frac{1}{2}\|\vec{w}\|^2 + C\sum_{i=1}^{l}\xi_i$$

C is a parameter that controls the tradeoff between margin and classification error ξ. The greater the value of C, the greater the penalty for errors, so the training process becomes tighter.

$$\underset{\alpha}{\text{Maximize}} \quad \sum_{i=1}^{l}\alpha_i - \frac{1}{2}\sum_{i,j=1}^{l}\alpha_i\alpha_j y_i y_j <\vec{x}_i\cdot\vec{x}_j> \tag{13}$$

$$\text{Subject to} \quad 0 \leq \alpha_i \leq C \quad (i = 1,2,\ldots,l) \qquad \sum_{i=1}^{l}\alpha_i y_i = 0 \tag{18}$$

Soft Margin

Based on the Karush-Kuhn-Tuchker complementary condition, solution (13) fulfills the following:

$$
\begin{array}{lcll}
\alpha_i = 0 & \Rightarrow & y_i f(x_i) > 1 & \\
0 < \alpha_i < C & \Rightarrow & y_i f(x_i) = 1 & \text{(unbounded SVs)} \\
\alpha_i = C & \Rightarrow & y_i f(x_i) < 1 & \text{(bounded SVs)}
\end{array}
\qquad (19)
$$

Cristianini-Taylor: Support Vector Machines and other kernel-based learning methods, Cambridge Univ.Press (2000) p.107
Vapnik, V. (1998): Statistical Learning Theory, Wiley, New York

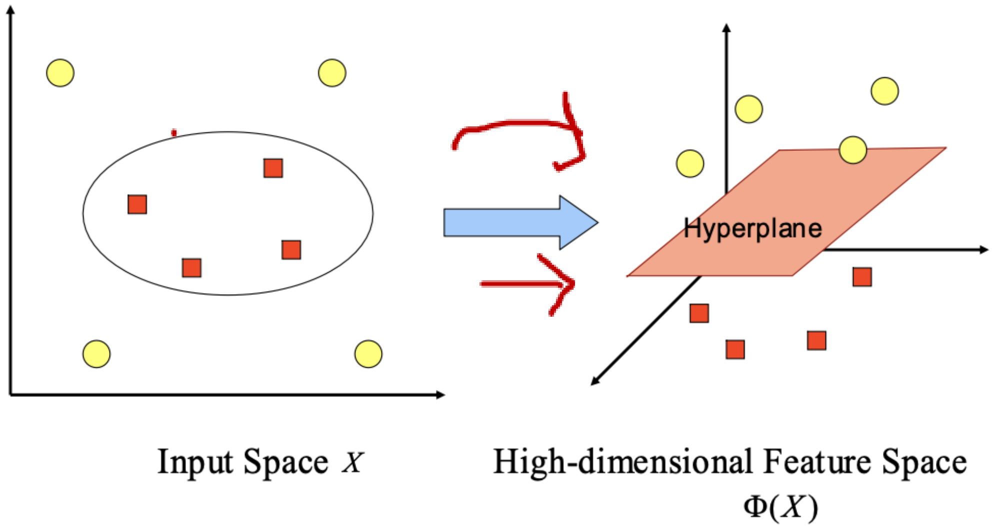
Parameter C is determined by trying several values and evaluating their effect on the accuracy achieved by SVM (e.g. by Cross-validation or by Design of Experiments - DOE).

Non-Linear Classification in SVM

- In general, problems in the real-world domain are rarely linear separable.
- Most are non-linear. To solve the non-linear problem, SVM is modified by including the **Kernel function**.
- In non-linear SVM, first the $\vec{x}$ data is mapped by the $\Phi(\vec{x})$ function to a higher-dimensional vector space. In this new vector space, a hyperplane that separates the two classes can be constructed.
- This is in line with Cover's theory which states "If a transformation is non-linear and the dimensions of the feature space are high enough, then the data in the input space can be mapped to a new feature space, where the patterns in high probability can be separated linearly".

Mapping to Feature Space

The Φ function maps data to a vector space with a higher dimension, so that the two classes can be separated linearly by a hyperplane.



The mathematical notation of this mapping is as follows.

$$\Phi : \Re^d \rightarrow \Re^q \qquad d < q$$

This mapping is done by maintaining the topology of the data, in the sense that two data that are closely spaced in the input space will also be closely spaced in the feature space, on the other hand, two data that are far away in the input space will also be far apart in the feature space.

Kernel Trick

- Furthermore, the learning process in SVM in finding support vector points only depends on the dot product of the data that has been transformed into a new higher-dimensional space: $\Phi(\vec{x}_i).\Phi(\vec{x}_j)$.
- Since the general transformation Φ is unknown, and it is very difficult to understand easily, the calculation of the dot product according to Mercer's theory can be replaced by a kernel function $K(\vec{x_i}, \vec{x_j})$ which implicitly defines the transformation of Φ.
- This is known as the Kernel Trick, which is formulated

$$K(\vec{x}_i, \vec{x}_j) = \Phi(\vec{x}_i).\Phi(\vec{x}_j)$$

- The kernel trick provides various conveniences, because in the SVM learning process, to determine the support vector, we only need to know the kernel function used, and we don't need to know the shape of the non-linear function .

Kernel Function

Various types of kernel functions are known, as summarized in the table below.

Polynomial	$K(\vec{x}_i, \vec{x}_j) = (\vec{x}_i . \vec{x}_j + 1)^p$
Gaussian	$K(\vec{x}_i, \vec{x}_j) = \exp(-\frac{\left\Vert \vec{x}_i - \vec{x}_j \right\Vert^2}{2\sigma^2})$
Sigmoid	$K(\vec{x}_i, \vec{x}_j) = \tanh(\alpha\vec{x}_i . \vec{x}_j + \beta)$

Non-Linear Classification in SVM

- SVM structure is a linear unit
- Non-linear classification is carried out in 2 stages
  1. The data is mapped from the original feature space $\Re^d$ to a new high-dimensional space $\Re^q$ using a non-linear function Φ , so that the data is distributed linearly separable. $\Phi : \Re^d \rightarrow \Re^q \quad d < q$
  2. Classification is carried out on the new space linearly.

- Using the Kernel Trick allows us to not need to explicitly compute the mapping function Φ

$$K(\vec{x}, \vec{x}') = < \phi(\vec{x}) \cdot \phi(\vec{x}') >$$

Non-Linear Classification in SVM

- Decision function on non linear classification:

$$f(\phi(\vec{x})) = < \vec{w} \cdot \phi(\vec{x}) > + b$$

- which can be written as the equation below.

$$\begin{aligned} f(\phi(\vec{x})) &= \sum_{i=1, x_i \in SV}^{n} \alpha_i y_i < \phi(\vec{x}) \cdot \phi(\vec{x}_i) > + b \\ &= \sum_{i=1, x_i \in SV}^{n} \alpha_i y_i K(\vec{x} \cdot \vec{x}_i) + b \end{aligned}$$

- The characteristics of the mapping function $\Phi(\vec{x})$ is difficult to analyze
- Kernel Trick uses $K(\overrightarrow{x_i}, \overrightarrow{x_i})$ instead of calculating $\Phi(\vec{x})$

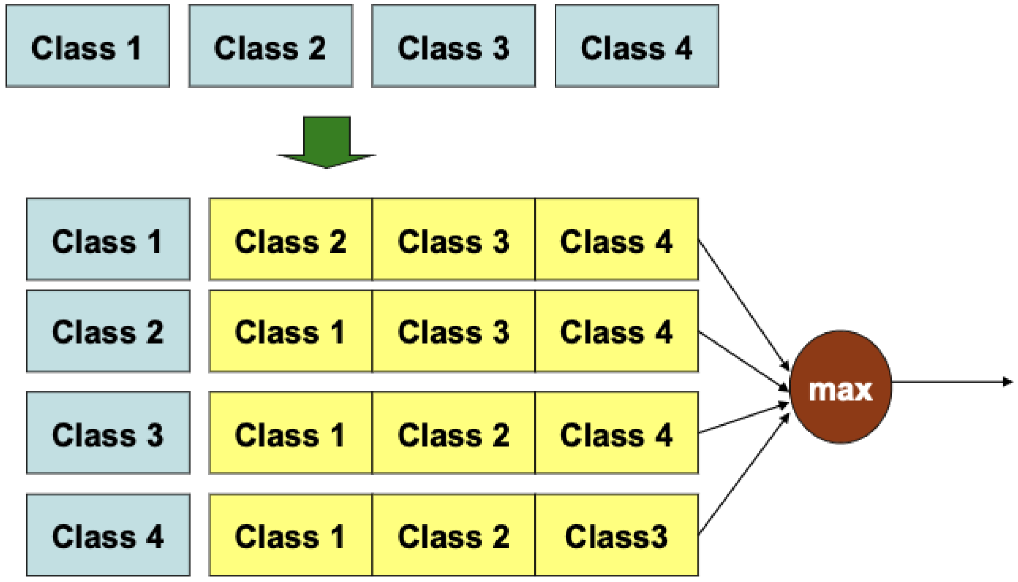
Sequential Method

- The optimal hyperplane in SVM can be found by formulating it into a Quadratic Programming problem and solved by a library that is widely available in numerical analysis.
- Another simple alternative is the sequential method developed by Vijayakumar, as follows.
    1. Initialization $\alpha_i = 0$
       Calculate the matrix $D_{ij} = y_i y_j \left(K\left(x_i, x_j\right) + \lambda^2\right)$
    2. Perform steps (a), (b), and (c) below for $i = 1, 2 \ldots, l$
        a. $E_i = \sum_{j=1}^{l} \alpha_j D_{ij}$
        b. $\delta\alpha_i = min\{max[\gamma(1 - E_i), -\alpha_i], C - \alpha_i\}$
        c. $\alpha_i = \alpha_i + \delta\alpha_i$
    3. Back to step 2 until the value of α converges (no significant change)
- In the above algorithm, γ is a parameter to control the speed of the learning process.
- Convergence can be defined from the level of change in the α value.

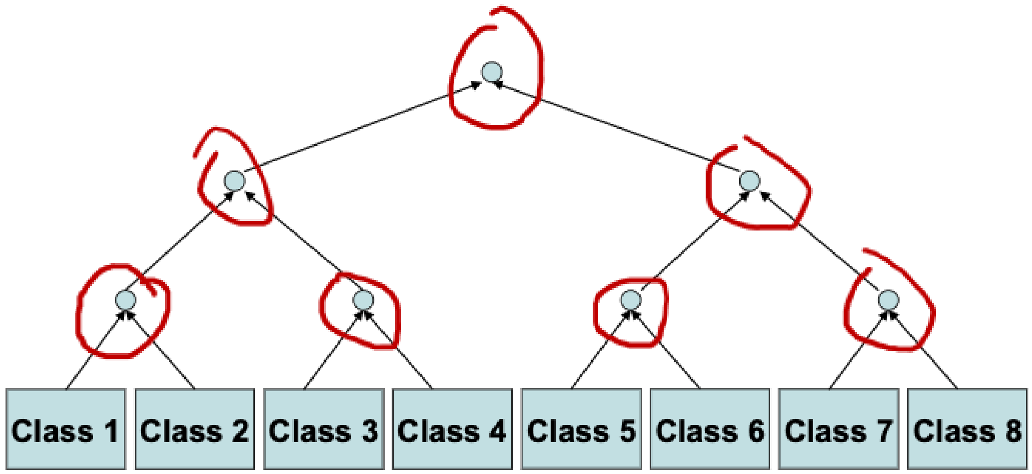
Multiclass Problem

- In principle, SVM is a binary classifier
- Expansion to multiclass classifier:
  - One vs Others Approach
  - One vs One : tree structured approach
    - Bottom-up tree (Pairwise)
    - Top-down tree (Decision Directed Acyclic Graph)
- In terms of training effort: One to Others is better than One vs One
- Runtime : both require evaluation of q SVMs (q = num. of classes)

One vs Others Approach

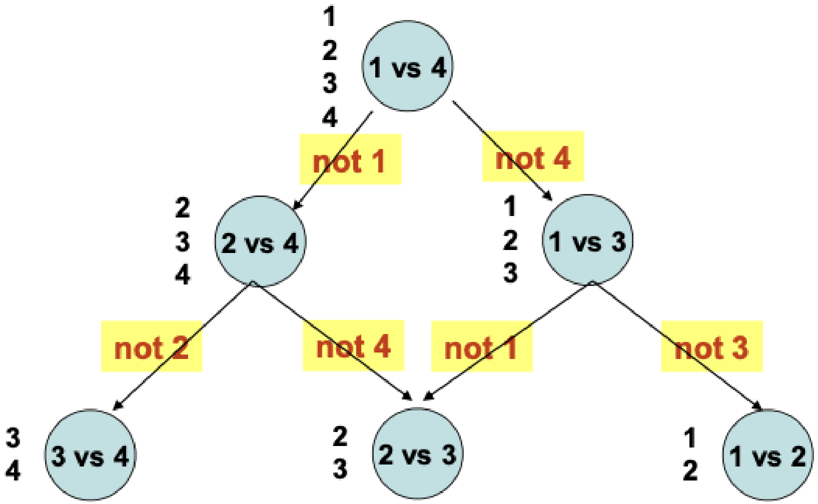


Bottom-up Tree



Proposed by Pontil and Verri

Top-Down Tree (DDAG)



Proposed by Platt et al.

SVM Characteristics

- In principle, SVM is a linear classifier.
- Pattern recognition is done by transforming the data in the input space to a higher-dimensional space, and optimization is done on the new vector space. This approach distinguishes SVM from pattern recognition solutions in general, which perform parameter optimization on the transformed result space with a lower dimension than the input space dimension.
- Implementing a Structural Risk Minimization (SRM) strategy.
- The working principle of SVM is basically only able to handle the classification of two classes.

The Advantages of SVM

- **Generalization**
  - Generalization is defined as the ability of a method to classify a pattern, which does not include the data used in the learning phase of that method.
  - Vapnik explained that the generalization error is influenced by two factors: the error on the training set, and another factor that is influenced by the Vapnik-Chervokinensis dimension.
  - Learning strategies on neural networks and generally machine learning methods are focused on efforts to minimize errors in the training set. This strategy is called Empirical Risk Minimization (ERM).
  - As for SVM, besides minimizing errors in the training set, it also minimizes the second factor. This strategy is called Structural Risk Minimization (SRM), and in SVM it is realized by selecting the hyperplane with the largest margin.
  - Various empirical studies show that the SRM approach to SVM provides a generalization error that is smaller than that obtained from the ERM strategy on neural networks and other methods.

The Advantages of SVM

- **Curse of Dimensionality**
  - Curse of dimensionality is defined as the problem faced by a pattern recognition method in estimating parameters (e.g. the number of hidden neurons in the neural network, stopping criteria in the learning process, etc.) due to the relatively small number of data samples compared to the dimensions of the data vector space.
  - The higher the dimensions of the vector space of the processed information, the consequence is the need for the amount of data in the learning process. In fact, it often happens that the amount of data is limited, and collecting more data is not possible due to cost constraints and technical difficulties. Under these conditions, if the method is "forced" to work on relatively small amounts of data compared to its dimensions, it will make the process of estimating the method parameters very difficult.
  - The curse of dimensionality is often experienced in biomedical engineering applications, because usually the available biological data is very limited, and its provision requires high costs.
  - Vapnik proved that the level of generalization obtained by SVM is not affected by the dimensions of the input vector. This is the reason why SVM is one of the appropriate methods used to solve high-dimensional problems, within the limited sample data available.

The Advantages of SVM

- **Theoretical basis**
  - As a statistical-based method, SVM has a theoretical basis that can be analyzed clearly and is not a black box.

- **Feasibility**
  - SVM can be implemented relatively easily because the process of determining the support vector can be formulated in the Quadratic Programming problem.
  - Thus, if we have a library to solve the QP problem, SVM can be implemented easily.
  - In addition, it can be solved by the sequential method as described previously.

The Limitation of SVM

- Difficult to use in large-scale problems. Large scale in this case is meant the number of samples processed.
- SVM is theoretically developed for classification problems with two classes. Currently, SVM has been modified to solve problems with more than two classes, including the One versus rest strategy and the Tree Structure strategy. However, each of these strategies has weaknesses, so it can be said that research and development of SVM on multiclass-problem is still an open research theme.